

## Questions:

Slide-1

- ① Draw diff. topology of computer network (Slide 10, 11)
- ② " a three layer protocol stack of postal mail service (Page - 15)
- ③ Draw a 3 layer protocol stack of showing logical connection / peer to peer communication (Stt Page - 21)
- ④ Mention 2 basic func. of Phy. Layer (29 → 2, 3 points)
- ⑤ Compare Simplex, half duplex, full duplex
- ⑥ Compare Error detection & error correction (P-32)
- ⑦ Define flow control (P-33 (1, 2) para)
- ⑧ Mention 3 basic func. of network layer (36)
- ⑨ Give the concepts of checkpoints of session Layer (P-45, first 2 part with diagram)
- ⑩ Give diff. formats of data used in presentation Layer (P-46, 1, 2 para)
- ⑪ Give Operating principle of hub. What is its limitation. (Slide 2, P-15).

## Questions

- ① Compare utilization of transmission line for smaller & larger frame under Stop and wait flow control. (Page 44) .

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## Slide 2 Questions

- ① What do you mean by frame at LLC Sublayer  
(Slide - 2, P-2)
- ② Draw byte stuffing & unstuffing technique
- ③ in char. oriented protocol (P-9 diagram)
- ④ Draw the idea of bit stuffing and unstuffing under flag bits with bit stuffing (P-12)
- ⑤ Compare the best case & worst case of bit stuffing & byte stuffing (P-13)
- ⑥ Compare error correction through retransmission and autonomous error correction (P-15, 15)
- ⑦ Example - 1 (P-22, 23)
- ⑧ How to avoid single bit error, 2 bit error, odd no. of bits in error, burst error (P-29-32)  
x31x
- ⑨ What will happen if  $E(x) = G(x)$  (P-33)
- ⑩ Give operative principal of sliding window flow control. (P-45)

10) Draw few steps of sliding window protocol for case of variable window size. (P: 50-52)

11) Compare Pure & slotted ALOHA (P: 63 upper) (no fig)

12) Show that max efficiency of pure ALOHA is 0.184 and slotted ALOHA is 0.3684.

(P: 68-71) (No: fig)

13) Example 1 (P: 72)

14) Compare Adv Exhaustive Service, Gated Service & Limited Service of token ring

protocol. (No fig, P: 78(a), 78(b), 79(c))

15) Why wireless network LAN use CSMA/CA instead of CD (P: 88)

(Binary Exp backoff 4M)

## Questions - Slide 4

- ① Give the basic idea of subnet (P: 6)
- ② How subnet is evaluated (P: 9)  $\begin{matrix} Q \rightarrow \text{Ans.} \\ \uparrow \quad \uparrow \end{matrix}$
- ③ Problem (Slide Page no: 13-15, 20, 21, 25)
- ④ What is CIDR (P: 39)
- ⑤ Problem (P: 43)
- ⑥ Give the concept of NAT with appropriate diagram (P: 46-47)
- ⑦ How IPv6 address is represented (P: 48)

### Questions - Slide 3

- ① Give an example of heterogenous network with activities of routers. (P: 3-4)  $\rightarrow$  (Diagram P4 only)
- ② What is the func<sup>n</sup> of differentiated service of IP header (P: 7, diagram x)
- ③                                  TTL (P: 12)
- ④ Give an example of calculation of header checksum (P: 15-17)

⑤ Give an example of tunneling of IP packet  
(P: 23-24, diagram 1 & 2)

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## Question on Lecture 5

- ① Compare connectionless and connection-oriented service (P-4)
- ② Example of service routing (P-6, diagram and table only)
- ③ Example of connection-oriented (P-9, fig. only)
- ④ Compare datagram, subnet, virtual circuit (P-11, only 3 diff.)
- ⑤ How fairness, optimally conflict each other show with example (P-14, first part with fig)
- ⑥ Problem on sink tree (P: 17-19)
- ⑦ How-pa Which parameters are considered in measuring path dist. in routing algo (P: 20, last part having only 6 parameters).

- 8) How <sup>excess</sup> flooding is protected in flooding algo. (P: 36, 3 points)
- 9) Problem on distance vector routing (P: 43-44)
- 10) How " " " responses to good news n  
bad new (P: 45-46  $\Delta$  कृष्ण कल्ले पिल्ले शिव)

## Questions on Lecture - 6

- i) Define Network congestion (P-2, first 2 para)
- ii) compare logical n physical " (P-4, upper para)
- iii) explain operations of load shedding to  
relieve relief congestion of datagram (P: 10-11)
- iv) take your decision about text and  
multimedia traffic data (P: 10-11)
- v) Define jitter
- vi) compare email, file transfer, video on demand  
and video conferencing in context of QoS  
parameter (P: 25 table)
- vii) Example (P: 30, 31)
- viii) problem on token bucket (P: 35-36)

L-12 → Online

L-12

## Question on Lecture-8

- ① What's the problem on 2 way handshaking in connection establishment under transport layer mention with example (P: 9-10)
- ② Draw the basic of 3 way handshaking (P: 11 (a) fig)
- ③ Old duplicate CR and old duplicate DR Data packet are rejected under 3 way hand shaking (P: 13 fig only)
- ④ Compare Symmetric & asymmetric release of connection (P: 14 second para)
- ⑤ Show the 3 way hand shake of connection release under TCP (P: 18 only diagram)
- ⑥ How it tracks long of DR (Disconnect request) (P: 20-24)
- ⑦ What is CWND and RWND (P: 30 of TCP (P: 34 1st part))
- ⑧ How ACK clock is acquired (P: 37-39) by a sender

L-14 Ques 20

L-13-16

L-15 Online

## Network Security (L-10)

- ① Define plaintext & Ciphertext with mathe. expression (P:4)
- ② Define cryptography, cryptology, (P:5 - last 3 lgs defi)
- ③ compare active & passive intruder (P:6)
- ④ give ex. of substitution cipher (P:12)
- ⑤ P:15 as example (P:15) Transposition cipher
- ⑥ compare substitution cipher & basic
- ⑦ principal ~~first~~ (1st diff. definition  
2nd: statistical distribution)
- ⑧ example of RSA algo (P:22-24)
- ⑨ compare conventional & digital sign (P:27-28)
- ⑩ Give the basic <sup>conditions</sup> concept of msg digest func (P:38-40)  
4 points)
- ⑪ Explain <sup>operation</sup> secure digital sign. under SHA1 algo (P:40-42)
- ⑫ Mention 3 examples of packet filtering in a firewall (P:55). What
- ⑬ What is firewall (P:52, first para)
- ⑭ How app gateway firewall works (P:57)
- ⑮ How intruder can enter through app gateway firewall (P:58)

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## List of Question on Application Layer (Lecture-9)

- ① Draw the arrange of socket in process to process communication (P: 4 only fig, no description)
- ② How a process is addressed in a host? (P: 5)
- ③ Give the idea of Net. App & App. protocol (P: 8, 2nd para)
- ④ Why DNS server is used? (P: 10)
- ⑤ Draw the hierarchy of Domain Name & label (P: 13 fig, no description)

⑥ Give the steps of DNS name resolution

(P: 21-22)

⑦ Mention the function of five tuple of DNS record (P: 254 - 25, no table)

⑧ Example - 1 (P: 26, no table, 1 line ans.)

⑨ Example - 2 (P: 27, table and after part)

⑩ How a mail box is addressed under SMTP (P: 33)

⑪ Mention the functions of 3 phrases of POP3 (P: 37)

⑫ State what are the benefits of web caching (P: 58)

⑬ Why Net. Mgt S/W is necessary (P: 70)